

Error Analysis

Method

Reviewed all 18 test-set inference results and cross-checked against the confusion matrix generated during validation.

Error Table

#	Image	Actual Objects	Model Prediction	Error Type	Possible Reason
1	-25_png.jpg	Person + Helmet	Only Helmet detected	False Negative (Person missed)	Person partially visible/occluded
2	-80_png.jpg	2 Persons + 2 Helmets	2 Persons, only 1 Helmet	False Negative (Helmet missed)	Small/overlapping helmet
3	img129_png.jpg	Person + without_helmet	Only without_helmet	False Negative (Person missed)	Person cropped or small in frame
4	img212_png.jpg	Person + without_helmet	Only without_helmet	False Negative (Person missed)	Low lighting / awkward angle
5	img263_png.jpg	Person + without_helmet	Only without_helmet	False Negative (Person missed)	Person partially out of frame
6	img276_png.jpg	Person + without_helmet	Only without_helmet	False Negative (Person missed)	Small person size in image
7	img38_png.jpg	Person + without_helmet	Only without_helmet	False Negative (Person missed)	Occlusion / low contrast
8	Confusion Matrix	Helmet	Predicted as background	False Negative	Helmet blended with background
9	Confusion Matrix	Without_helmet	Predicted as background	False Negative	Weak head/hair contrast
10	Confusion Matrix	Person (6 cases)	Predicted as background	False Negative	Person only partially visible

Summary of Error Types

- False Negatives (Person missed): The most common error is the model sometimes fails to draw the broader "person" box even when it correctly detects the helmet/without_helmet region.
- False Negatives (Helmet/without_helmet as background): A smaller number of boxes were missed entirely, mostly linked to small object size or blending with the background.
- No major wrong-class confusions were found (e.g. helmet confused for without_helmet), suggesting the two classes are visually distinct in this dataset.

Suggested Improvements

- Add more training images for the "person" class, especially with partial/occluded bodies, to reduce person false negatives.
- Increase training epochs (e.g. 50-60) to let the model converge further, since the loss curves were still gently decreasing at epoch 30.
- Use a larger base model (YOLOv8s instead of YOLOv8n) for higher capacity to capture small or overlapping objects.
- Balance the dataset across classes — ensure roughly equal numbers of helmet, without_helmet, and person instances.
- Add low-light and occluded examples deliberately to the training set to improve robustness in those conditions.